ASSIGNMENT – 1

PCCST501 – Computer Networks

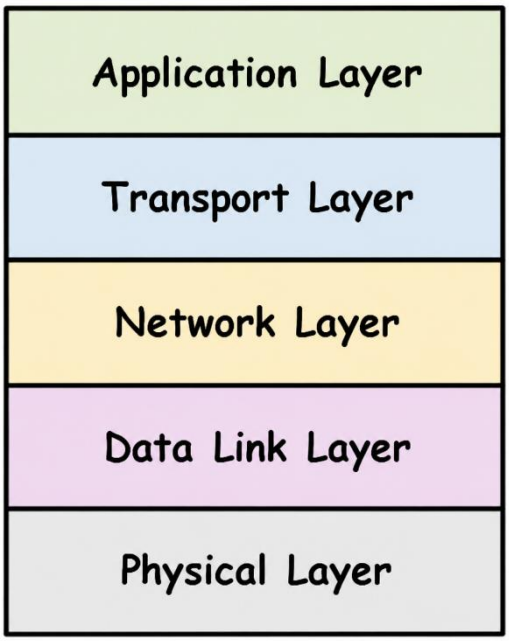
How the Internet Delivers a Web Page

Website Chosen: https://www.google.com

Q1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer.

The TCP/IP protocol suite is the communication model used for data exchange over the Internet. It divides communication into four layers, with each layer performing a specific function. These layers work together to ensure reliable and efficient transmission of data between devices.

TCP/IP Protocol Stack:



Explanation of Each Layer

1. Application Layer

The Application Layer is the top layer of the TCP/IP protocol stack. It provides services directly to users and applications. Protocols like HTTP, HTTPS, DNS, FTP, and SMTP work at this layer. It is used for web browsing, email, and file transfer.

2. Transport Layer

The Transport Layer is responsible for reliable communication between devices. It uses TCP for reliable data transfer and UDP for faster communication. This layer ensures that data reaches the destination correctly.

3. Network Layer

The Network Layer handles the routing of data packets using IP addresses. It selects the best path for sending data from the source to the destination across different networks.

4. Data Link Layer

The Data Link Layer transfers data between devices on the same network. It converts packets into frames, uses MAC addresses for communication, and detects transmission errors.

5. Physical Layer

The Physical Layer is responsible for transmitting data through physical media such as Ethernet cables, Wi-Fi, or fiber optics. It sends and receives data in the form of electrical, optical, or wireless signals.

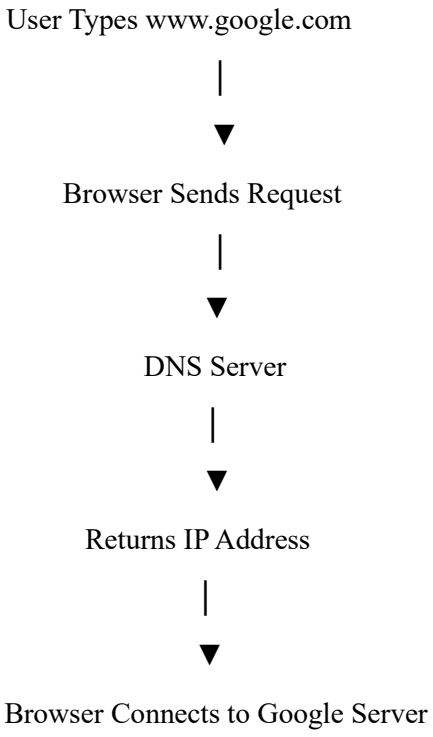
Q2. Explain the Role of DNS in Converting a Domain Name into an IP Address.

The Domain Name System (DNS) is an Application Layer protocol that converts a human-readable domain name into its corresponding IP address. Computers communicate using IP addresses, whereas users prefer domain names because they are easier to remember.

For example, instead of remembering a numerical IP address, users simply type:

www.google.com

The DNS server finds the matching IP address and returns it to the browser.



Q3. Describe how HTTP is used to request and receive the webpage.

HTTP (HyperText Transfer Protocol) is an Application Layer protocol used for communication between a web browser (client) and a web server. Today, most websites use **HTTPS**, which provides secure communication.

Steps Involved

1. The user enters **www.google.com** in the web browser.
2. The browser sends an **HTTP/HTTPS request** to Google's web server.
3. The server processes the request.
4. The server sends an **HTTP response** containing the webpage.
5. The browser receives the response and displays the Google homepage

Q4. Explain how TCP provides reliable communication for HTTP.

TCP (Transmission Control Protocol) is a connection-oriented protocol that works at the Transport Layer of the TCP/IP protocol suite. It provides reliable communication between the client and the server. Before data transfer begins, TCP establishes a connection between the sender and the receiver.

TCP divides the data into small segments and sends them over the network. It checks whether each segment reaches the destination correctly. If any segment is lost or damaged during transmission, TCP automatically retransmits it. It also delivers the data in the correct order, ensuring that the complete webpage is received without errors.

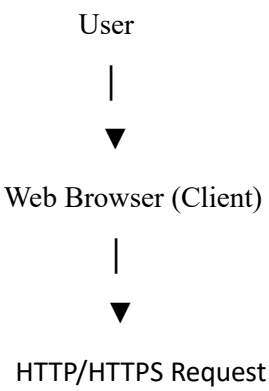
Because of these features, HTTP uses TCP to provide reliable communication between the browser and the web server.

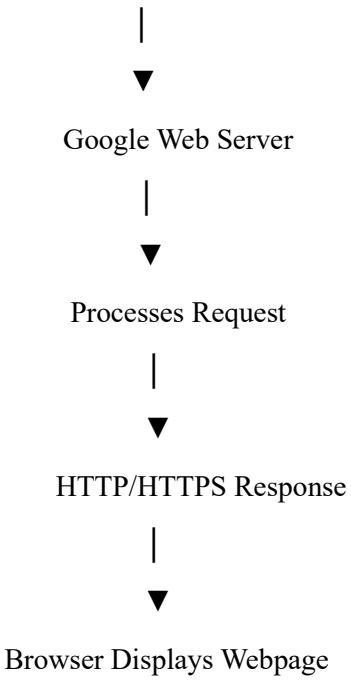
Q5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

The **Google website** follows the **Client–Server architecture**. In this architecture, the **web browser acts as the client** and the **Google web server acts as the server**. The client sends a request for a webpage, and the server receives the request, processes it, and sends the required webpage back to the client.

In Google, all the data and web pages are stored on the server. Users access this information by sending requests through their browsers. The clients do not communicate directly with each other; instead, they communicate only with the server. This makes the Client–Server architecture reliable, efficient, and suitable for web applications.

Client–Server Flow Diagram





Q6.Compare HTTP and FTP with at least four differences.

HTTP	FTP
Used to access web pages.	Used to transfer files.
Default Port: 80 (HTTPS uses 443).	Default Port: 21.
Mainly transfers HTML, CSS, and images.	Transfers files and folders.
Used through web browsers.	Used through FTP clients.
Stateless protocol.	Maintains a file transfer session.

Q7. Explain how cookies improve user experience with one real-world example.

Cookies are small text files stored by the web browser on the user's device. They help websites remember user information and preferences.

Cookies improve the user experience by remembering login details, language settings, and user preferences. They also help websites provide personalized content.

Real-World Example

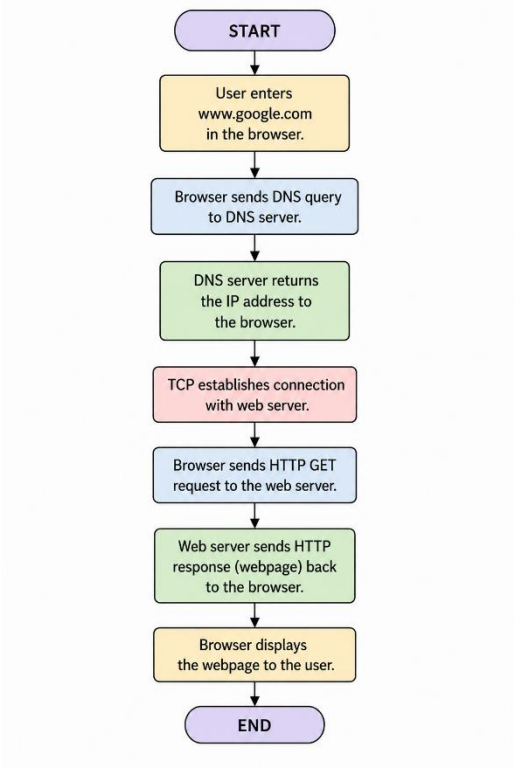
When a user logs in to a Google account, Google stores a cookie in the browser. The next time the user visits Google or Gmail, the browser sends the cookie to Google's server. This allows the user to remain logged in without entering the username and password again.

Advantages of Cookies

- Remember user login.

- Save user preferences.
- Provide personalized content.
- Improve browsing experience.

Q8. Draw a flow diagram showing the complete communication from the browser to the web server and back.



Q9. Conclusion

The Internet works efficiently because different application layer protocols perform specific tasks. When a user enters a website address, **DNS** converts the domain name into an IP address, **TCP** establishes a reliable connection between the client and the server, and **HTTP/HTTPS** is used to request and receive the webpage. These protocols work together to ensure that information is transferred accurately and quickly. **Cookies** further improve the user experience by remembering login details and user preferences. Understanding these protocols helps us learn how web communication takes place and why networking is important in our daily lives. Application layer protocols make Internet services such as web browsing, email, and file transfer simple, reliable, and efficient for users.